NAME	REGISTER NO.	ASSESSMENT
MIRUDHULA DAISY M	9923005207	Mini Assessment - Prompt Workflow Design

AI CAREER & PLACEMENT PLANNER

Multi-Step Prompt Workflow for Personalized Placement Preparation

01 | Concept

Purpose: AI Career & Placement Planner converts a student's academic background, skills, career goals and preparation level into a personalized placement preparation strategy.

Problem: Students often have multiple skills to improve, different job roles to target and limited preparation time. They may not know which skills to prioritize, what to study first, or how to prepare for interviews. A chained AI prompt workflow transforms these inputs into a structured placement preparation plan.

02 | Objective

The workflow uses sequential prompts to understand the student's academic background, identify skill gaps, prioritize preparation areas, create a realistic placement preparation schedule, generate practice questions and review progress to update the remaining plan. Each stage uses relevant information produced by earlier stages without writing code.

03 | Workflow Overview

INPUT Student Details → 1 Profile Analysis → 2 Skill Gap → 3 Prioritize → 4 Schedule → 5 Practice → 6 Review → **OUTPUT** Personalized Placement Plan

04 | Workflow Steps

1. Student Profile Analysis — Analyze academic background, technical skills, career goal and preparation level. **Output:** Student Profile + Strengths & Weaknesses

2. Skill Gap Identification — Compare current skills with target job requirements. **Output:** Skill-Gap Analysis

3. Preparation Prioritization — Rank preparation areas using importance, difficulty and urgency. **Output:** Prioritized Preparation List

4. Preparation Schedule — Create a realistic day-by-day preparation plan. **Output:** Placement Preparation Timetable

5. Practice & Revision — Generate aptitude, technical, coding and interview questions. **Output:** Practice Set + Answer Key

6. Progress Review — Review progress and adapt the remaining preparation plan. **Output:** Updated Preparation Plan

05 | Prompt Workflow (UI Execution Prompts)

Step 1 - Extract Student Profile

Analyze the student's academic background, technical skills, career goal, current preparation level and available study time. Identify strengths and weak areas. Do not invent missing information. Create a concise career profile.

Step 2 - Analyze Skill Gaps

Using the Step 1 student profile, compare the student's current skills with the requirements of the target job role. Identify skills that are strong, need improvement, or are missing.

Step 3 - Prioritize Preparation

Using the Step 2 skill-gap analysis, rank preparation areas based on importance, difficulty, confidence level and placement urgency. Explain the reason for each priority.

Step 4 - Build Preparation Schedule

Using the Step 3 prioritized list, create a realistic day-by-day placement preparation schedule. Include technical preparation, aptitude, coding, interview preparation, revision, breaks and buffer time.

Step 5 - Generate Practice

Using the Step 4 preparation schedule, generate suitable aptitude, technical, coding and interview practice questions. Include a short answer key and focus on important concepts.

Step 6 - Review & Adapt

Review the student's progress against the preparation plan. Identify completed topics, weak areas and unfinished tasks. Adapt the remaining schedule and explain what changed and why.

06 | Prompt Chaining Architecture

Step 1 establishes the student profile; Step 2 uses it to identify skill gaps; Step 3 uses those gaps to prioritize preparation; Step 4 uses the priorities to create the timetable; Step 5 uses the timetable to generate practice; and Step 6 uses progress information to update the remaining plan.

STAGE	INPUT / CONTEXT	OUTPUT
1	Student details	Student profile
2	Step 1 profile + job requirements	Skill gaps
3	Step 2 skill gaps	Priority list
4	Step 3 priorities + available time	Timetable
5	Step 4 timetable + topics	Practice set
6	Progress + earlier context	Updated plan

07 | Demonstration Input

Candidate: Final-year ECE student with a CGPA of 8.4. The student knows Java and SQL, wants to prepare for campus placements, has 10 days remaining, can study 4 hours per day, and has low confidence in aptitude and medium confidence in coding and communication.

08 | Expected Final Output

A personalized placement preparation package containing a prioritized skill list, day-by-day timetable, practice questions, progress review and an updated plan based on the student's performance.

09 | Workflow Result & Evaluation

The completed workflow transforms student information into a synchronized placement preparation roadmap. It replaces guesswork with guided prompt chaining. Evaluation focuses on correct use of each prompt stage, meaningful use of output.

10 | Sample Final Preparation Plan (Day-by-Day Schedule)

TIMELINE	FOCUS AREA	PLANNED ACTIVITY	DELIVERABLE / OUTPUT
Day 1	Aptitude	Basics + practice questions	10 questions
Day 2	Java	Core concepts + coding	5 programs
Day 3	SQL	Queries + practice	Query set
Day 4	Technical	ECE concepts + revision	Short notes
Day 5	Interview	HR + technical questions	Mock interview
Day 6	Revision	Mock test + weak areas	Readiness score

11 | Key Benefits

- Personalized preparation based on the student's skills and goals
- Priority-based learning to focus on important skill gaps
- Time-efficient scheduling with a realistic study plan
- Practice and revision through AI-generated questions
- Progress-based improvement by updating the plan
- No coding required for designing the workflow

12 | Conclusion

AI Career & Placement Planner demonstrates how prompt engineering can divide a complex placement preparation task into connected stages. By combining profile analysis, skill-gap identification, prioritization, scheduling, practice and progress review, the workflow produces a personalized and adaptable preparation plan for students.